

Music Recommendation System

comprehensive Project Report: Million Song Dataset Analysis

Big Data Mining

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Abstract

This project develops a robust music recommendation system using the Million Song Dataset (10,000-song subset). To address the "Echo Chamber Effect" prevalent in traditional similarity-based systems, we implemented and evaluated five distinct algorithms, including Maximal Marginal Relevance (MMR) and Truncated SVD Matrix Factorization. A unified evaluation framework was established to assess Relevance, Diversity, Novelty, and Coverage. Experimental results demonstrate that the MMR algorithm achieves the optimal balance, significantly improving recommendation diversity by 31.7% while maintaining high relevance, yielding a weighted composite score of 0.8484. Additionally, memory optimization techniques using Truncated SVD reduced memory consumption by 87%.

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1 Introduction

1.1 Research Motivation

In the era of streaming, recommendation systems are pivotal. However, traditional systems often suffer from the "Echo Chamber Effect," where users are trapped in a loop of highly similar content. This project aims to break this cycle by developing a system that balances accuracy with serendipity.

1.2 Objectives

- **Algorithm Implementation:** Develop Content-Based (Cosine), Diversity-Optimized (MMR/MMR+Temp), and Latent Factor (SVD) models.
- **Evaluation Framework:** Assess performance across four dimensions: Relevance, Diversity, Novelty, and Coverage.
- **Optimization:** Implement memory-efficient techniques to handle high-dimensional feature matrices.

2 System Environment Setup

To ensure reproducibility and performance, the system environment is configured as follows:

2.1 Hardware Requirements

- **RAM:** Minimum 8 GB (16 GB Recommended).
- **Storage:** Approx. 2 GB for dataset and output logs.
- **Processor:** Multi-core CPU recommended for matrix operations.

2.2 Software Stack

The project is built using **Python 3.x** with the following core libraries:

- **NumPy & Pandas:** For numerical computing and data manipulation.
- **H5py:** For extracting data from HDF5 binary files.
- **SciPy:** Specifically `scipy.sparse.linalg` for Truncated SVD decomposition.
- **Matplotlib & Seaborn:** For generating visualization dashboards.

3 Research Technical Methods

3.1 Feature Engineering

We utilized the *Timbre* segments from the Million Song Dataset. For each song, the raw audio analysis is converted into a fixed-length feature vector:

$$Feature_Vector = [\mu_1, \dots, \mu_{12}, \sigma_1, \dots, \sigma_{12}] \quad (1)$$

Where μ_i represents the mean and σ_i represents the standard deviation of the i -th timbre dimension. This results in a 24-dimensional vector representing the acoustic texture of the track.

3.2 Recommendation Algorithms

3.2.1 Cosine Similarity (Baseline)

Calculates the normalized dot product between song vectors. It maximizes relevance but ignores diversity.

3.2.2 Maximal Marginal Relevance (MMR)

To improve diversity, MMR iteratively selects items that maximize the following objective function:

$$MMR = \arg \max_{d_i \in R \setminus S} [\lambda \cdot Sim_1(d_i, Q) - (1 - \lambda) \cdot \max_{d_j \in S} Sim_2(d_i, d_j)] \quad (2)$$

Where:

- $\lambda = 0.7$: The weight balancing Relevance (Sim_1) and Diversity penalty (Sim_2).
- Q : Query song; S : Set of already selected songs.

3.2.3 Stochastic MMR (Gumbel-Max)

We introduced a Temperature parameter ($T = 0.5$) and Gumbel noise to the MMR score to allow for controlled exploration (serendipity) rather than purely deterministic ranking.

3.2.4 Matrix Factorization (Truncated SVD)

To handle the $10,000 \times 10,000$ similarity matrix efficiently, we applied Truncated SVD:

$$S \approx U_k \Sigma_k V_k^T \quad (3)$$

By keeping only the top $k = 50$ singular values, we reduced memory usage by **87%** (from 763 MiB to < 100 MiB) while retaining 100% of the explained variance for the latent factors.

4 System Usage

The system is designed as a modular pipeline. Follow these steps to execute the project:

1. **Data Preparation:** Place the Million Song Subset HDF5 files into the `data/MillionSongSubset` directory.
2. **Execution:** Run the main Jupyter Notebook (`new_version.ipynb`). The pipeline will automatically:
 - Extract features from all 10,000 songs.
 - Compute the similarity matrix.
 - Run all 5 recommendation algorithms on the test set.

5 Experimental Results

The system was evaluated using a 90/10 Train-Test split (1,000 test songs).

5.1 Performance Metrics

We established a Unified Evaluation Framework covering:

- **Relevance:** Cosine similarity to the query.
- **Diversity:** Intra-List Distance (ILD).
- **Novelty:** Ability to recommend long-tail items.
- **Coverage:** Percentage of the catalog recommended.

5.2 Comparative Analysis

The comprehensive results are summarized below:

Method	Relevance	Diversity	Novelty	Coverage	Weighted Score
Cosine (Base)	0.9736	0.0359	0.9986	36.91%	0.4854
MMR (Best)	0.9717	0.0472	0.9986	38.82%	0.8484
MMR+Temp	0.9595	0.0529	0.9985	36.56%	0.3500
MF (SVD)	0.9644	0.0401	0.9986	38.52%	0.4874

Table 1: Performance Comparison Table (Weighted Score: Div 35%, Rel 25%, Nov 25%, Cov 15%)

5.3 Key Findings

1. **MMR Dominance:** The MMR algorithm achieved the highest composite score. It improved Diversity by **31.7%** compared to the baseline, with only a negligible 0.2% drop in Relevance.
2. **Exploration Trade-off:** The Stochastic MMR (MMR+Temp) yielded the highest diversity (+47.5%) but sacrificed 1.45% in relevance, proving useful for "Shuffle" or "Discovery" modes.
3. **SVD Efficiency:** While slightly less accurate than pure Cosine, the Truncated SVD method demonstrated excellent scalability and discovered non-linear latent relationships.

6 Conclusion and Future Work

6.1 Conclusion

This project successfully implemented a scalable, multi-algorithm music recommendation system. By integrating Maximal Marginal Relevance (MMR) and unified evaluation metrics, we proved that it is possible to mitigate the "Echo Chamber Effect" without compromising user satisfaction (Relevance). The memory-optimized SVD approach further ensures the system can scale to larger datasets.

6.2 Future Work

- **Hybridization:** Incorporate user listening history (Collaborative Filtering) to solve the "cold start" problem for new songs.
- **Deep Learning:** Replace statistical feature extraction with Neural Collaborative Filtering (NCF) or Autoencoders for better feature representation.
- **Online Learning:** Implement Contextual Bandits to dynamically adjust the diversity weight (λ) based on real-time user feedback.

References

- [1] Bertin-Mahieux, T., et al. "The Million Song Dataset." ISMIR 2011.
- [2] Carbonell, J., & Goldstein, J. "The use of MMR, diversity-based reranking." SIGIR 1998.
- [3] Koren, Y., et al. "Matrix factorization techniques for recommender systems." Computer 42.8 (2009).